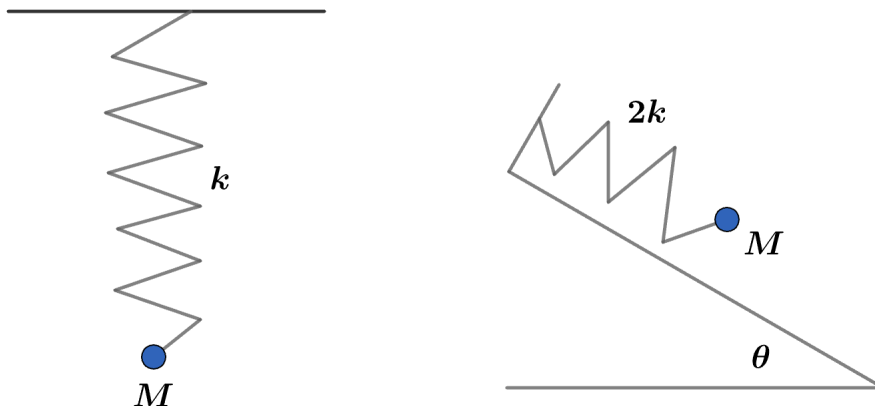


2014 F=ma Exam: Problem 8

Kevin S. Huang



From the period formula for a mass-spring system, we have

$$T_0 = 2\pi\sqrt{\frac{M}{k}}$$

By cutting the spring in half, the spring constant gets doubled since adding two half-springs in parallel should result in the original spring,

$$\begin{aligned}\frac{1}{k_{\text{half}}} + \frac{1}{k_{\text{half}}} &= \frac{2}{k_{\text{half}}} = \frac{1}{k} \\ k_{\text{half}} &= 2k\end{aligned}$$

Oscillating on an inclined plane doesn't affect the period since it is independent of g . Thus, the new period is

$$T' = 2\pi\sqrt{\frac{M}{k_{\text{half}}}} = 2\pi\sqrt{\frac{M}{2k}} = \frac{T_0}{\sqrt{2}}$$

so the answer is D.